

Hashing - II

Hash functions

1. Division method

$$h(k) = k \bmod m$$

given key

of slots in the table

Caveats: m should not have a small divisor

extreme example: If m is a power of 2, i.e., $m = 2^r$
then the hash does not depend on all bits of k .

if $k = 1011000111\underbrace{011010}_2$, $r = 6$

then $h(k) = 011010$

Doesn't depend on all bits of k !

fix: Pick m to be a prime, not too close to a power of 2.

2. Multiplication method

- Assume all keys are integers,
- $m = 2^r$, and
- Computer has w -bit word

Define the hash function as:

$$h(k) = (A \cdot k \bmod 2^w) \gg (w-r)$$

Here, A is an odd integer in range $2^{w-1} < A < 2^w$

Example: $m = 8 = 2^3$, let $w = 7$ -bit

$$\begin{array}{r} \bullet 1011001 = A \end{array}$$

$$\begin{array}{r} x \quad 1101011 = K \end{array}$$

$$\begin{array}{r} \hline 1001010 \bullet 0110011 \\ \hline \end{array}$$

most sig. 3-bits = right shift by $(w-r)$

Resolving collisions: Open Addressing

- No storage for links
- Insertions systematically probe the table until an empty slot is found
- The hash function takes two arguments:
the key and a probe number

$$h: U \times \{0, 1, \dots, m-1\} \rightarrow \{0, 1, \dots, m-1\}$$

Universal set of keys Probe # Slot no (or index to T)

- For a given key k , the probe sequence $\langle h(k, 0), h(k, 1), \dots, h(k, m-1) \rangle$ should be a permutation of $\langle 0, 1, \dots, m-1 \rangle$
- $n \leq m$
- Deletion is difficult (not impossible though)

- Search uses the same probe seq.

↳ Successful if k is found

↳ Unsuccessful if NIL is found

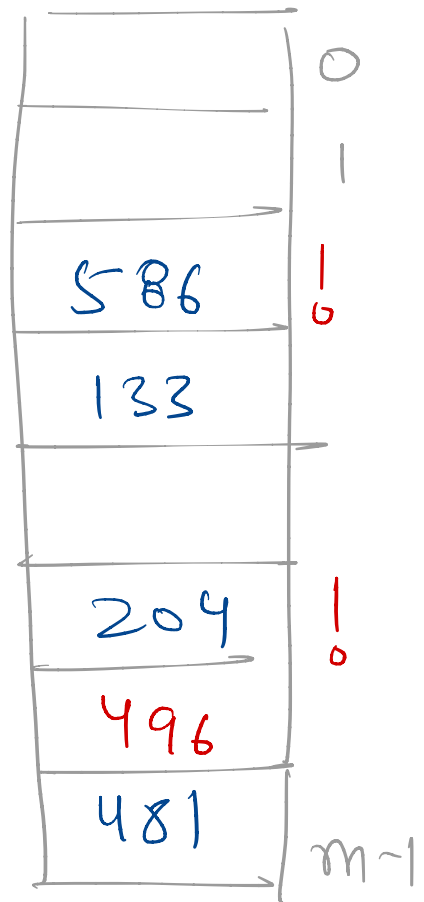
Example: Insertion with open addressing

Insert key $k = 496$

0. Probe $h(496, 0)$

2. Probe $h(496, 1)$

3. Probe $h(496, 2)$



- while searching go through the same probe sequence.

Hash-Insert (T, k)

$i = 0$

repeat

$q = h(k, i)$

 if $T[q] == \text{NIL}$

$T[q] = k$

 return q

 else $i = i + 1$

until $i == m$

error "hash table overflow"

Hash-Search (T, k)

$i = 0$

repeat

$q = h(k, i)$

 if $T[q] == k$

 return q

 else $i = i + 1$

until $T[q] == \text{NIL}$ or $i == m$

return NIL

Probing Strategies

Linear probing:

$$h(k, i) = (h'(k) + i) \bmod m$$

↗ Probe #

↙ ordinary hash function

- It suffers from primary clustering.
Where long runs of occupied slots build up & tend to get longer.
- Results in longer search time.

Double hashing

— Given two ordinary hash functions $h_1(k)$ and $h_2(k)$, double hashing uses:

$$h(k, i) = (h_1(k) + i \cdot h_2(k)) \bmod m$$

— Usually pick $m = 2^r$, and $h_2(k)$ odd

— In order for the entire table to be searched, $h_2(k)$ must be relatively prime to the hash table size

Two possibilities:

1. let $m = 2^r$ & $h_2(k)$ produce odd numbers

2. m be prime and $h_2(k)$ always returns a positive integer less than m .

$$h_1(k) = k \bmod m$$

$$h_2(k) = 1 + (k \bmod m')$$

where m' is slightly less than m (e.g. $m' = m - 1$)

Theorem: Given an open addressed hash table, with load factor $\alpha = n/m < 1$, the expected number of probes in an unsuccessful search is at most $\frac{1}{(1-\alpha)}$

$$E[\# \text{ probes}] \leq \frac{1}{(1-\alpha)} \quad \text{load factor}$$

Proof:

- At least one probe is necessary
- with probability n/m , a collision happens.
 $\Rightarrow 2^{\text{nd}}$ probe is needed
- with probability $(n-1)/(m-1)$, a collision happens,
 $\Rightarrow 3^{\text{rd}}$ probe is needed

⋮

Observe: $\frac{n-i}{m-i} < \frac{n}{m}$ for $i = 1, 2, \dots, n$

Therefore, the expected # of probes is

$$= 1 + \frac{n}{m} \left(1 + \frac{n-1}{m-1} \left(1 + \frac{n-2}{m-2} \left(\dots \left(1 + \frac{1}{m-n+1} \right) \dots \right) \right) \right)$$

$$\leq 1 + \alpha (1 + \alpha (1 + \alpha (\dots (1 + \alpha) \dots)))$$

$$\leq 1 + \alpha + \alpha^2 + \dots +$$

$$= \sum_{i=0}^{\infty} \alpha^i = \frac{1}{(1-\alpha)} \quad \text{upper bound.}$$

- If α is const., then accessing open addressed table takes const. time

- If table is half full, the avg. no. of probes for unsuccess. search = 2

$$\left(= \frac{1}{(1-0.5)} \right)$$

- If table is 90% full, avg # of probes = 10

- If table is full (i.e., $\alpha=1$), then all m slots must be probed.